

Activity 2

Prepare an album of UPVC doors and windows.

2.6 Aluminium

Aluminium is the world's most abundant metal and is the third most common element, comprising 8% of the earth's crust. But it is commercially produced mainly from bauxite which is hydrated oxide of Aluminium. The versatility of Aluminium makes it the most widely used metal after steel.







2.6.1 Properties of Aluminium

1. It is a good conductor of heat and electricity.
2. It is rarely attacked by nitric acid, organic acid or water.
3. It is highly resistant to corrosion.
4. It is light in weight, malleable and ductile.
5. It is a soft metal.
6. The melting point of aluminium is about 658°C.
7. It possesses great toughness and tensile strength(124-290 N/mm^2).
8. Its specific gravity is about 2.70.

2.6.2 Uses of Aluminium as Building Material

Aluminum is used as an important building material especially in developing countries. The following are the important properties of aluminum that make it useful as a building material.

1. **Air tightness:** A well-designed aluminium door, window, etc., is perfectly airtight and sealed for dust and rain water.
2. **Appearance:** The finished aluminium will be having a pleasing appearance and depending on the decorative style of the building, shades of various colours can be selected.
3. **Ease in fabrication and assembly:** As aluminium is comparatively soft and ductile, the fabrication of doors, windows, etc., can be easily carried out. It can be easily dismantled, transported and re-erected.
4. **Handling and transport cost:** Aluminium is very light and hence the handling and transport cost is very low.

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